



COVENANT UNIVERSITY

CANAANLAND, KM 10, IDIROKO ROAD
P.M.B 1023, OTA, OGUN STATE, NIGERIA.

TITLE OF EXAMINATION: B.Sc EXAMINATION

COLLEGE: SCIENCE AND TECHNOLOGY

SCHOOL: NATURAL AND APPLIED SCIENCES

DEPARTMENT: COMPUTER & INFORMATION SCIENCES

SESSION: 2015/2016

SEMESTER: ALPHA

COURSE CODE: CSC 214

CREDIT

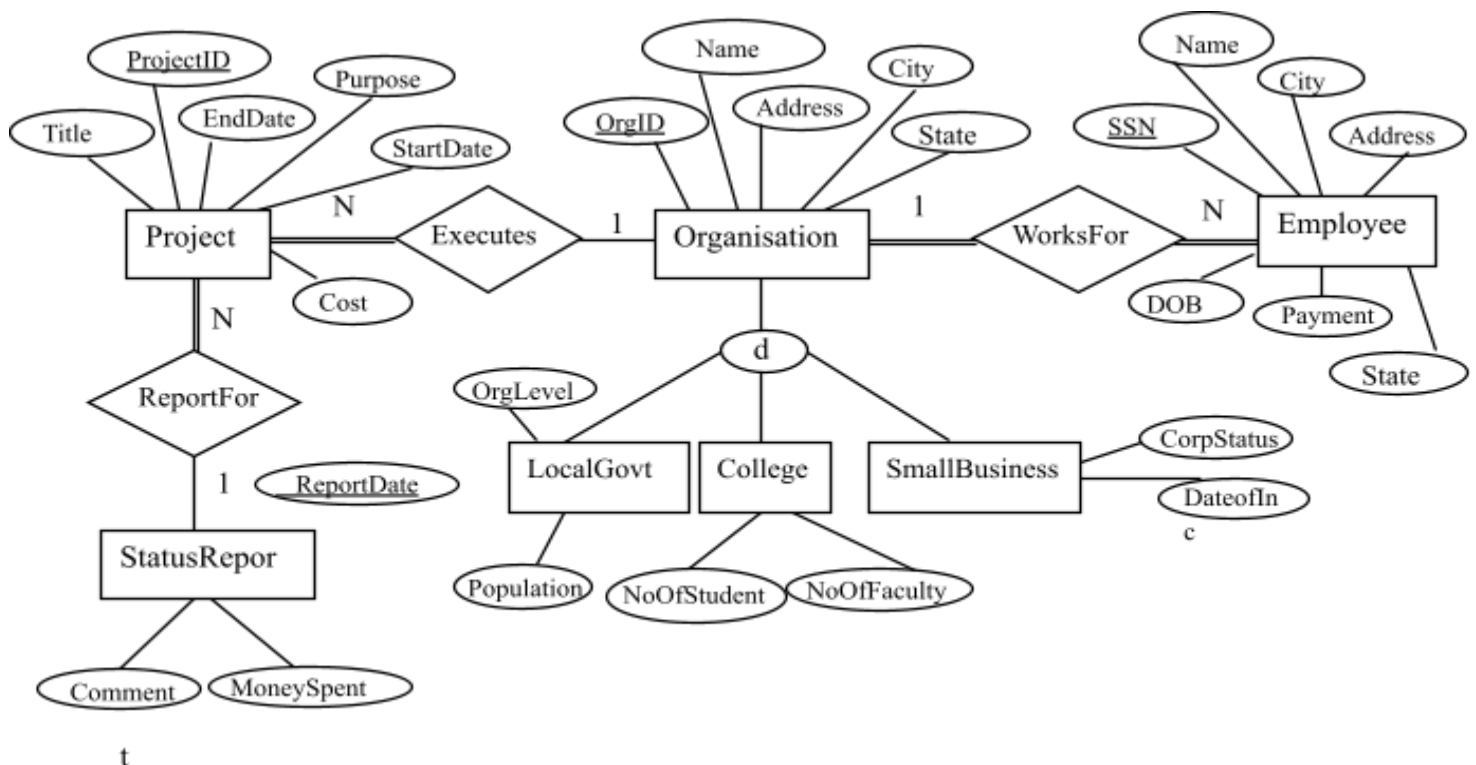
UNIT: 3

COURSE TITLE: HIGH PERFORMANCE AND DATABASE MANAGEMENT SYSTEM

INSTRUCTION: ANSWER QUESTION ONE AND ANY OTHER THREE QUESTIONS

TIME: 3HOURS

- 1(a) State in maximum of 10 short sentences the step by step process of connecting SQL Server Database Management System to Visual Studio programming languages environment. (5marks)
- 1(b) Show the Functional dependency diagram for Table 1: Patient _Appointment Table in question (3) (5marks)
- 1(c) Explain five limitation of file-Based Approach (5marks)
- 1(d) Convert the following ER diagram into a set of tables. (10marks)



2(a) What is the relationship between Candidate key and Foreign keys (5marks)

2 (b) Use the following business rules to create the ERD for a small consulting company named ConsultCo. Show all **entities, relationships, optionalities, connectivities and cardinalities**.

A client can contract many projects, but each project is contracted by one client. Some clients in the company client list have not (yet) contracted any project(s). A project may have many (consulting company) employees assigned to it, and each employee may be assigned to many projects. Some employees are not assigned to any project. One employee leads each project. An employee may lead more than one project at a time. One employee coordinates each project. An employee may coordinate more than one project at a time. Some employees never lead or coordinate projects. Many of the consulting company employees have multiple job skills that are applicable to the consulting work. Each job description- system analyst, database designer, accountant and so on- can fit many employees. Some job descriptions are not yet filled. Some employees do not have a specific job classification. 10marks)

3 . Given the Patient _Appointment Table below

- (i) Determine the candidate keys for the table (3marks)
- (ii) Normalizing the table below to Boyce-Codd Normal form and Show the final decomposed relations with the data. (12marks)

Table 1: Patient _Appointment Table

Functional dependency

PNo, ADate, $\rightarrow$ StaffNo , ATime, DentistName, SurgeonNo (pk)

PNo $\rightarrow$ PName (pd)

StaffNo $\rightarrow$ DentistName Td

StaffNo , ADate $\rightarrow$ SurgeonNo

Date, ATime, SurgeonNo $\rightarrow$ PNo , PName, StaffNo, DentistName (ck)

StaffNo, ADate, ATime $\rightarrow$ PNo , PName (ck)

4. (a) Discuss each of the following concepts in the context of the relational data model

- (i) attribute and tuple
- (ii) domain
- (iii) degree and cardinality. (5marks)

- (b) *Hotel* (hotelNo, hotelName, City)
Room (roomNo, hotelNo, type, price)
Booking (hotelNo, guestNo, dateFrom, dateTo, roomNo)
Guest (questNo, guestName, questAddress)

Using the schema above to answer the following questions:

- (i) Identify the foreign keys in this schema (2marks)
- (ii) Explain how the entity integrity and referential integrity rules apply to these relations (4marks)
- (iii) Produce some sample tables for these relations that observe the relational integrity. (4marks)

5. (a) Define the 3 basic relational algebra operation (4marks)

- (a) Using the schema in 4(c), generate the relational Algebra for the following queries:

- (i) List all hotels
- (ii) List all single rooms with a price below N2000 per night
- (iii) List the names and Address of all guests

(3marks)

- (b) Using the schema in 4(c), generate SQL statement for the following queries:

(8marks)

- (i) How many hotels are there?
- (ii) List full details of all hotels in London
- (iii) List number of rooms in each hotel
- (iv) List the names and address of all guests living in London, alphabetically ordered by name. use pattern matching operator.

- 6(a) What is data mining? (3marks)

- (b) With diagram representation show data mining as the core component of knowledge discovery process. (3marks)

- (c) What is concurrency control (3marks)

- (d) Explain some of the potential problems caused by concurrency (4.5marks)

- (e) What is dead lock? (1.5marks)